STD 1: Measurement (Std 1)

M2 Working with Time (Y11)

Teacher: Denis Vlismas

Exam Equivalent Time: 57 minutes *(based on HSC allocation of 1.5 minutes approx. per mark)*

TOPIC OVERVIEW

- *MS-M2 Working with Time* attracted a solitary 2-mark longer answer question in the *2020 Std1 exam* which followed a solitary multiple choice question in 2019.
- *MS-M2 Working with Time* content has been significantly diminished when compared to the old Gen2 course, although some of this lost content is again examinable from 2021 onwards (details below).
- A number of past HSC questions retained in the database have been adjusted, omitting content no longer examinable, to reflect the current syllabus.

ANALYSIS - Common pitfalls

- *Working With Time* was examined in the *2020 Std1 exam* with a standard "time differences" question that produced a surprisingly low mean mark of 28%. Revision attention to this question style is recommended.
- Exposure to Universal Coordinated Time (UTC) and similar terminology is an important feature of database questions.
- Applicable to the *2021 Std1 exam* onwards, students are now required to "understand the link between longitude and time to find time differences". This change is reflected in the database.
- Accordingly, the database now includes examples that require the application of 15° = 1 hour, or 1° = 4 minutes time difference *(2017 Std2 27d, M2 SM-Bank 3)*.
- Some distance between locations (or "arc length" questions) have been adjusted from past HSC questions and included. We regard this as a possible cross topic application within measurement, provided the earth's radius is explicitly stated in the question.

Questions

1. Measurement, STD2 M2 SM-Bank 27 MC

Part of a train timetable is shown.

New Castle	13.17	13.28	...	13.58
Hamilton	13.23	...	14.00	14.07
Waratah	...	14.03	...	14.16
Blue Haven	14.01	14.16	14.23	14.28
Jeb's Bush	14.22	14.32	...	14.44
Gosford	14.36	...	14.51	14.58

Kalyn arrives at New Castle station at 1.25 pm and needs to get to Gosford as quickly as possible.

Assuming all trains run to schedule, what is the EARLIEST time that Kalyn can arrive at Gosford station?

A. 2.32 pm

B. 2.36 pm

C. 2.51 pm

D. 2.58 pm

2. Measurement, STD2 M2 2011 HSC 3 MC

Perth in Western Australia is 8 hours ahead of Coordinated Universal Time (UTC). Cape Town in South Africa is 2 hours ahead of UTC.

What is the time in Cape Town when it is 1 pm in Perth?

(A) 3 am

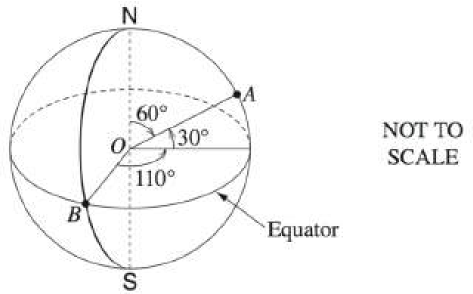
(B) 7 am

(C) 7 pm

(D) 11 pm

3. Measurement, STD2 M2 2010 HSC 15 MC

In this diagram of the Earth, O represents the centre and B lies on both the Equator and the Greenwich Meridan.



What is the latitude and longitude of point A?

(A) 30°N 110°E

(B) 30°N 110°W

(C) 60°N 110°E

(D) 60°N 110°W

---

4. Measurement, STD2 M2 2007 HSC 20 MC

Kim lives in Perth. He wants to watch an ice hockey game being played in Toronto starting at 10.00 pm on Wednesday.

Toronto is 13 hours behind Perth.

What is the time in Perth when the game starts?

(A) 9.00 am on Wednesday

(B) 7.40 pm on Wednesday

(C) 12.20 am on Thursday

(D) 11.00 am on Thursday

---

5. Measurement, STD1 M2 2019 HSC 2 MC

What is the time difference between 8:35 am and 2:10 pm?

A. 5 hours and 25 minutes

B. 5 hours and 35 minutes

C. 6 hours and 25 minutes

D. 6 hours and 35 minutes

---

6. Measurement, STD2 M2 SM-Bank 2

Bert is in Moscow, which is three hours behind of Coordinated Universal Time (UTC).

Karen is in Sydney, which is eleven hours ahead of UTC.

a. Bert is going to ring Karen at 9 pm on Tuesday, Moscow time. What day and time will it be in Sydney when he rings? ***(1 mark)***

b. Karen is going to fly from Sydney to Moscow. Her flight will leave on Wednesday at 8 am, Sydney time, and will take 15 hours. What day and time will it be in Moscow when she arrives? ***(2 marks)***

---

7. Measurement, STD2 M2 2013 HSC 27e

Karin is in Athens, which is two hours ahead of Greenwich Mean Time. Marco is in New York, which is five hours behind Greenwich Mean Time.

i. Karin is going to ring Marco at 10 pm on Tuesday, Athens time.
What day and time will it be in New York when she rings? ***(1 mark)***

ii. Marco is going to fly from New York to Athens. His flight will leave on Wednesday at 9 am, New York time, and will take 11 hours.
What day and time will it be in Athens when he arrives? ***(2 marks)***

---

8. Measurement, STD2 M2 2006 HSC 26d

Cassie flew from London to Manila. Manila is 8 hours ahead of London.

Her plane left London at 9.30 am Monday (London time), stopped for 5 hours in Singapore and arrived in Manila at 4.00 pm Tuesday (Manila time).

What was the total flying time? (Ignore time zones.) ***(2 marks)***

---

9. Measurement, STD2 M2 2016 HSC 27e

Melbourne time is 6 hours ahead of Dubai time.

A plane leaves Melbourne on Friday at 11.30 pm. The flight time to Dubai is 15 hours.

What will be the time and the day in Dubai when the plane is due to land? ***(2 marks)***

---

10. Measurement, STD2 M2 2018 HSC 29a

The time in Brisbane is 4.5 hours ahead of the time in New Delhi. John flew from New Delhi to Brisbane via Singapore. His plane left New Delhi at 11.30 am (New Delhi time), stopped for 3 hours in Singapore, and arrived in Brisbane at 9.00 am the following day (Brisbane time).

What was the plane's total flying time? ***(3 marks)***

---

11. Measurement, STD2 M2 SM-Bank 3

An aircraft travels at an average speed of 913 km/h. It departs from a town in Kenya (0°, 38°E) on Tuesday at 10 pm and flies east to a town in Borneo (0°, 113°E).

i. Calculate the distance (d), to the nearest kilometre, between the two towns, using

$$d = \frac{\theta}{360} \times 2\pi r \text{ where } \theta = 75^\circ \text{ and } r = 6400 \text{ km}$$ ***(2 marks)***

ii. How long will the flight take? (Answer to the nearest hour.) ***(1 mark)***

iii. What will be the day and local time in Borneo when the aircraft arrives? (Ignore time zones.) ***(2 marks)***

---

12. Measurement, STD2 M2 SM-Bank 4

Lee wants to call home to Sydney (34°S, 151°E) from Pateete, Tahiti (17°S, 149°W) when it is 7 pm on Friday in Sydney.

Sydney is eleven hours ahead of UTC. Papeete is ten hours behind UTC.

Give the time and day in Papeete when she should make the call. (Ignore time zones.) ***(2 marks)***

---

13. Measurement, STD2 M2 2009 HSC 26b

John lives in Denver and wants to ring a friend in Osaka.

i. In Denver it is 9 pm Monday. Given Osaka has a UTC of +9 and Denver has a UTC of –7, what time and day is it in Osaka then? ***(1 mark)***

ii. John's friend in Osaka sent him a text message which happened to take 14 hours to reach him. It was sent at 10 am Thursday, Osaka time.
What was the time and day in Denver when John received the text? ***(2 marks)***

---

14. Measurement, STD2 M2 2011 HSC 27b

Pontianak has a longitude of 109°E, and Jarvis Island has a longitude of 160°W.

Both places lie on the Equator.

i. Calculate the shortest distance between these two places (d), to the nearest kilometre, using

$$d = \frac{\theta}{360} \times 2\pi r \text{ where } \theta = 91^\circ \text{ and } r = 6400 \text{ km}$$ ***(1 mark)***

ii. The position of Rabaul is 4° to the south and 48° to the west of Jarvis Island. What is the latitude and longitude of Rabaul? ***(2 marks)***

---

15. Measurement, STD1 M2 2020 HSC 15

The time in Melbourne is 11 hours ahead of Coordinated Universal Time (UTC). The time in Honolulu is 10 hours behind UTC. A plane departs from Melbourne at 7 pm on Tuesday and lands in Honolulu 9 hours later.

What is the time and day in Honolulu when the plane lands? ***(2 marks)***

---

16. Measurement, STD1 M2 2021 HSC 15

City *A* is in Sweden and is located at (58°N, 16°E). Sydney, in Australia, is located at (33°S, 151°E).

Robert lives in Sydney and needs to give an online presentation to his colleagues in City *A* starting at 5:00 pm Thursday, local time in Sweden.

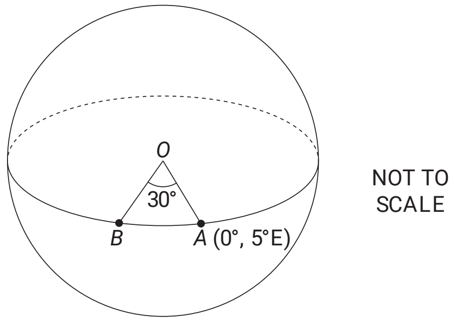
What time and day, in Sydney, should Robert start his presentation?

It is given that 15° = 1 hour time difference. Ignore daylight saving. ***(3 marks)***

---

17. Measurement, STD2 M2 2017 HSC 27d

Island *A* and island *B* are both on the equator. Island *B* is west of island *A*. The longitude of island *A* is 5°E and the angle at the centre of Earth (*O*), between *A* and *B*, is 30°.



a. What is the latitude and longitude of island B? ***(2 marks)***

b. What time is it on island B when it is 10 am on island A? ***(1 mark)***

Worked Solutions

1. Measurement, STD2 M2 SM-Bank 27 MC

1st train - Kalyn arrives at Newcastle late
2nd train - doesn't stop at Gosford
3rd train - arrives 2.51 pm
4th train - arrives 2.58 pm
$\Rightarrow$ C

2. Measurement, STD2 M2 2011 HSC 3 MC

Perth is 6 hrs ahead of Cape Town
$\therefore$ 1pm in Perth = 7 am in Cape Town
$\Rightarrow B$

3. Measurement, STD2 M2 2010 HSC 15 MC

Since A is 30° North of the Equator
$\Rightarrow$ Latitude is $30^\circ N$
Since A is 110° East of Greenwich
$\Rightarrow$ Longitude is $110^\circ E$

$\therefore A$ is $30^\circ N\ 110^\circ E$
$\Rightarrow A$

4. Measurement, STD2 M2 2007 HSC 20 MC

$\therefore$ Time in Perth
$= 10$ pm (Wed) $+ 13$ hours
$= 11$ am on Thursday
$\Rightarrow D$

5. Measurement, STD1 M2 2019 HSC 2 MC

8:35 am − 2:10 pm = 5 hours and 25 minutes
$\Rightarrow A$

♦ *Mean mark 48%.*

Worked Solutions

6. Measurement, STD2 M2 SM-Bank 2

a. Bert is 14 hours behind Karen.
At 9 pm in Moscow:

Time in Sydney = 9 pm + 14 hours
= 11 am Wednesday

b. Arrival time = 8 am + 15 hours
= 11 pm Wednesday (Sydney time)

$\therefore$ Time in Moscow = 11 pm less 14 hours
= 9 am Wednesday

7. Measurement, STD2 M2 2013 HSC 27e

i. Athens is Greenwich + 2 hours
New York is Greenwich - 5 hours
Athens is 7hrs ahead of New York
$\Rightarrow$ 10pm Tuesday in Athens = 3pm Tuesday in New York

$\therefore$ It will be 3pm Tuesday when Karin rings

ii. Flight takes 11 hours
If Marco leaves on Wed at 9am
$\Rightarrow$ He arrives at 9am + 11 hrs + 7 hours

$\therefore$ Marco arrives on Thurs at 3am

8. Measurement, STD2 M2 2006 HSC 26d

$\therefore$ Flight time = 9:30 (Mon) to 4:00 pm (Tues)
− (stopover + time difference)
$= 14.5 + 16 - (5 + 8)$
$= 17.5$ hours

9. Measurement, STD2 M2 2016 HSC 27e

$$\begin{aligned}\text{Arrival time} &= 11:30 + 15 \text{ hours}\\ &= 14:30 \text{ (Melbourne time)}\end{aligned}$$

$$\begin{aligned}&\therefore \text{ Arrival time in Dubai time}\\ &\quad = 14:30 - 6 \text{ hours}\\ &\quad = 8:30 \text{ am on Saturday}\end{aligned}$$

10. Measurement, STD2 M2 2018 HSC 29a

$$\text{11:30 am in New Delhi} = \text{4 pm in Brisbane}$$

$$\begin{aligned}\text{Total travel time} &= \text{4 pm} \rightarrow \text{9 am (next day)}\\ &= 17 \text{ hours}\end{aligned}$$

$$\begin{aligned}\therefore \text{ Flying time } &= 17 - 3\\ &= 14 \text{ hours}\end{aligned}$$

11. Measurement, STD2 M2 SM-Bank 3

i. Angular difference in longitude

$$\begin{aligned}&= 113 - 38\\ &= 75^\circ\end{aligned}$$

$$\begin{aligned}\therefore \text{ Distance } &= \frac{75}{360} \times 2 \times \pi \times 6400\\ &= 8377.58\ldots\\ &= 8378 \text{ km (nearest km)}\end{aligned}$$

ii. $$\begin{aligned}\text{Flight time } &= \frac{\text{Distance}}{\text{Speed}}\\ &= \frac{8378}{913}\\ &= 9.176\ldots\\ &= 9 \text{ hours (nearest hr)}\end{aligned}$$

iii. $$\begin{aligned}\text{Time Difference } &= 75 \times 4\\ &= 300 \text{ minutes}\\ &= 5 \text{ hours}\end{aligned}$$

Kenya is further East

$\Rightarrow$ Kenya is +5 hours

COMMENT: NESA Nov-19 syllabus updates ... use longitude to calculate time differences (1° = 4 mins).

$$\begin{aligned}&\therefore \text{ Arrival time in Kenya}\\ &\quad = \text{10 pm (Tues)} + \text{5 hrs} + \text{9 hrs (flight)}\\ &\quad = \text{12 midday on Wednesday}\end{aligned}$$

12. Measurement, STD2 M2 SM-Bank 4

Sydney is 21 hours ahead.

$$\begin{aligned}&\therefore \text{ Time in Papeete}\\ &\quad = \text{7pm (Fri) less 21 hours}\\ &\quad = \text{10pm (Thurs)}\end{aligned}$$

13. Measurement, STD2 M2 2009 HSC 26b

i. Denver is behind Osaka time by 16 hours.

$\therefore$ Time in Osaka $= 9$ pm Monday plus 16 hours
$= 1$ pm Tuesday

ii. Denver is 16 hours behind Osaka

♦ *Mean mark 34%*

$\therefore$ John will receive the text at 10 am Thursday less 16 hours plus 14 hours.
(i.e. 8 am Thursday.)

---

14. Measurement, STD2 M2 2011 HSC 27b

i.
$$\begin{aligned}\text{Shortest distance} &= \frac{91}{360} \times 2\pi r \\ &= \frac{91}{360} \times 2 \times \pi \times 6400 \\ &= 10\ 164.79... \\ &= 10\ 165 \text{ km (nearest km)}\end{aligned}$$

ii. Latitude

♦♦ *Mean mark 33%*

4° South of Jarvis Island
Since Jarvis Island is on equator
$\Rightarrow$ Latitude is 4°S

Longitude
Jarvis Island is 160°W
Rubail is 48° West of Jarvis Island, or 208° West
which is 28° past meridian (180°)

$$\begin{aligned}\Rightarrow \text{Longitude} &= (180 - 28)^\circ\text{E} \\ &= 152^\circ\text{E}\end{aligned}$$

$\therefore$ Position is (4°S, 152°E)

---

15. Measurement, STD1 M2 2020 HSC 15

Melbourne $\rightarrow$ UTC $+ 11$
Honolulu $\rightarrow$ UTC $- 10$
$\Rightarrow$ Honolulu is 21 hours behind Melbourne

♦♦ *Mean mark 28%.*

$\therefore$ Plane landing time
$= 7$ pm $+ 9$ hours
$= 4$ am (Wednesday – Melb)
$= 7$ am (Tuesday – Honolulu)

STRATEGY: 21 hours behind = 1 day behind + 3 hours.

---

16. Measurement, STD1 M2 2021 HSC 15

Angular difference $= 151 - 16 = 135^\circ$

$$\Rightarrow \text{Time difference} = \frac{135}{15} = 9 \text{ hours}$$

♦♦ *Mean mark 25%.*

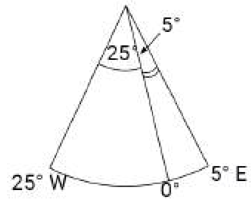
Sydney is east of Sweden $\rightarrow$ ahead

Presentation time (Sydney) $=$ 5 pm Thurs $+$ 9 hours
$=$ 2 am Friday

---

17. Measurement, STD2 M2 2017 HSC 27d

a.
$$\begin{aligned}\text{Longitude (island } B) &= 5 - 30 \\ &= -25 \\ &= 25^\circ \text{ W}\end{aligned}$$



$\therefore$ Island B is $(0^\circ, 25^\circ\text{W})$.

b. Time difference $= 30 \times 4 = 120$ mins $= 2$ hours
Since B is west of A,
Time on island $B = 10$ am less 2 hours
$= 8$ am

♦ *Mean marks of 40% and 45% for parts (i) and (ii) respectively.*
COMMENT: NESA Nov-19 syllabus updates ... use longitude to calculate time differences.